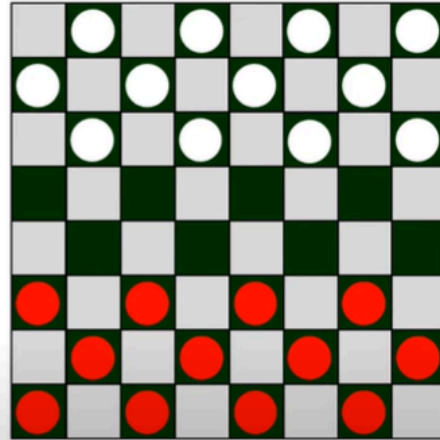


What is Machine Learning

Machine learning

"Field of study that gives computers the ability to learn without being explicitly programmed."

Arthur Samuel (1959)



If the checkers program had been allowed to play only ten games (instead of tens of thousands) against itself, a much smaller number of games, how would this have affected its performance?

- ☐ Would have made it better
- ☒ Would have made it worse

1. What is Machine Learning (ML)?

Machine Learning is a branch of **artificial intelligence (AI)** where computers learn from data to make predictions or decisions **without being explicitly programmed** for every single scenario.

Instead of writing fixed rules (**if this → then that**), we give the computer **examples** (data) and let it figure out the patterns on its own.

Simple Analogy

Think of training a dog:

- Instead of programming it step-by-step ("If you hear the word 'sit,' bend legs"), you show examples, reward correct actions, and over time, it learns.
 - ML works the same way — it learns patterns from examples.
-

Core Steps in Machine Learning

1. **Collect Data** → Numbers, text, images, sensor readings, etc.
 2. **Train Model** → Feed the data to an algorithm so it can find patterns.
 3. **Test Model** → Check if the model makes good predictions on new data.
 4. **Deploy Model** → Use it for real-world decisions.
-

Practical Example (General ML)

- **Problem:** Predict house prices.
- **Data:** Historical records with **size**, **location**, **bedrooms** → **price**.
- **Learning:** The ML model figures out how each factor influences price.
- **Prediction:** You give a new house's details, and it predicts the price.

Machine learning algorithms

rapid advancements

used most in real-world applications

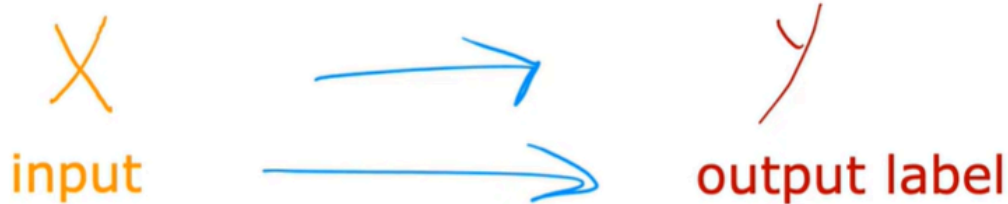
- Supervised learning ← course 1, 2
- Unsupervised learning ← course 3
- Recommender systems
- Reinforcement learning

Practical advice for applying learning algorithms



What is Supervised Learning

Supervised learning



Learns from being given "right answers"

2. What is Supervised Learning?

Definition:

Supervised learning is a type of ML where we train the model using **labeled data** — that means each training example has **both**:

- **Input** (features)
- **Correct output** (label/target)

The model “learns” the mapping from inputs to outputs.

Why “Supervised”?

It’s like having a teacher:

- The teacher gives you questions **and the correct answers** during training.
- You learn to generalize so you can answer new, unseen questions later.

Two Main Types of Supervised Learning

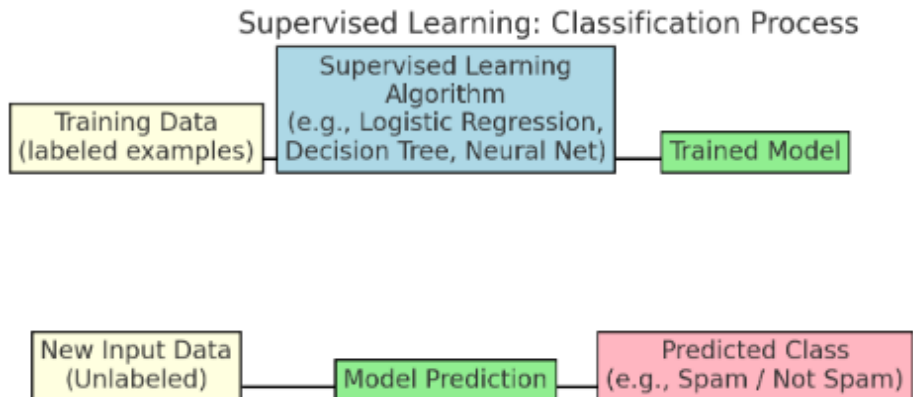
1. **Classification** → Predict categories.
 - Example: Email spam filter (`spam` / `not spam`).
 2. **Regression** → Predict continuous values.
 - Example: Predicting stock price or temperature.
-

Practical Example 1 (Classification)

Problem: Spam email detection

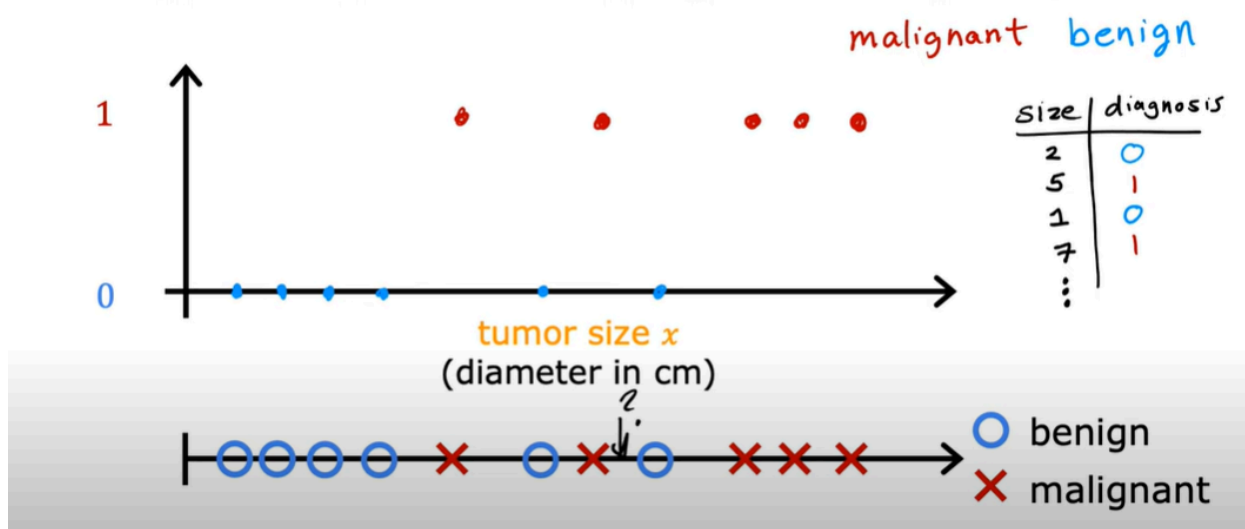
- **Input:** Email text, sender address, number of links, etc.
- **Label:** Spam (1) or Not Spam (0)
- **Algorithm:** Logistic Regression, Decision Trees, Neural Networks, etc.
- **Result:** New email comes in → Model predicts if it’s spam.

Classification :

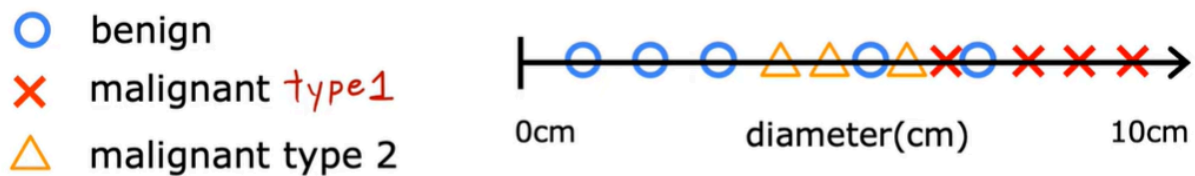


Input (X)	Output (Y)	Application
email	spam? (0/1)	spam filtering
audio	text transcripts	speech recognition
English	Spanish	machine translation
ad, user info	click? (0/1)	online advertising
image, radar info	position of other cars	self-driving car
image of phone	defect? (0/1)	visual inspection

Classification: Breast cancer detection



Classification: Breast cancer detection



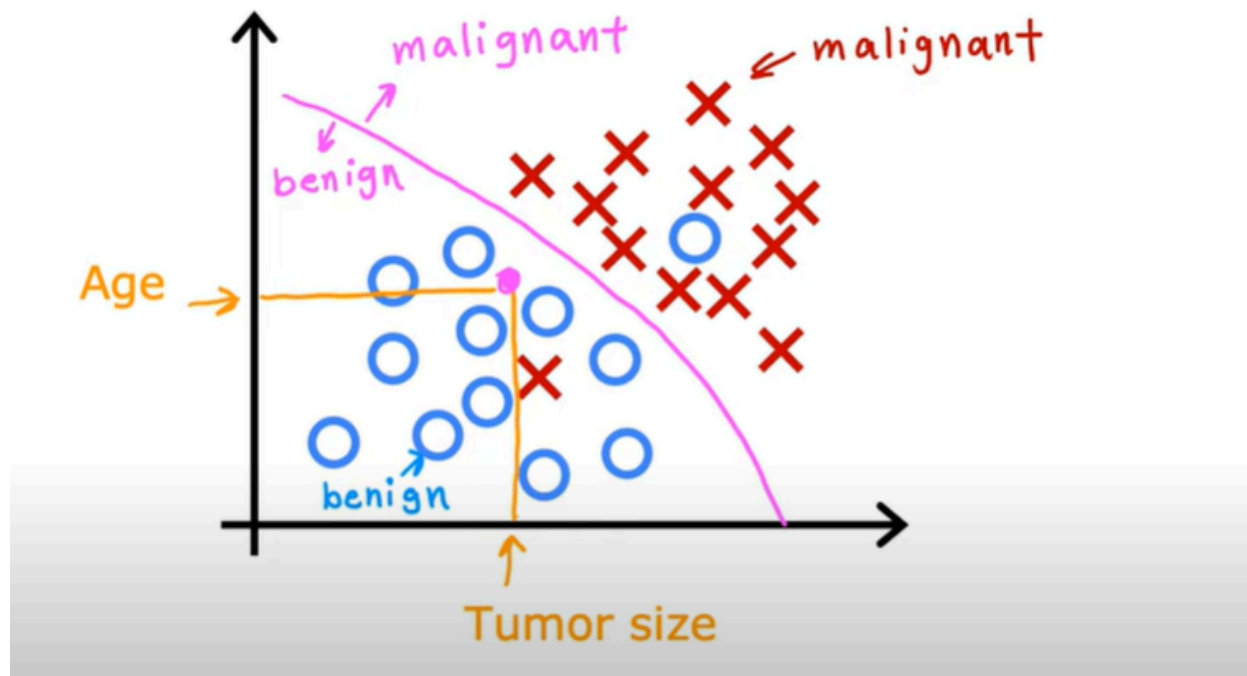
class category

Classification

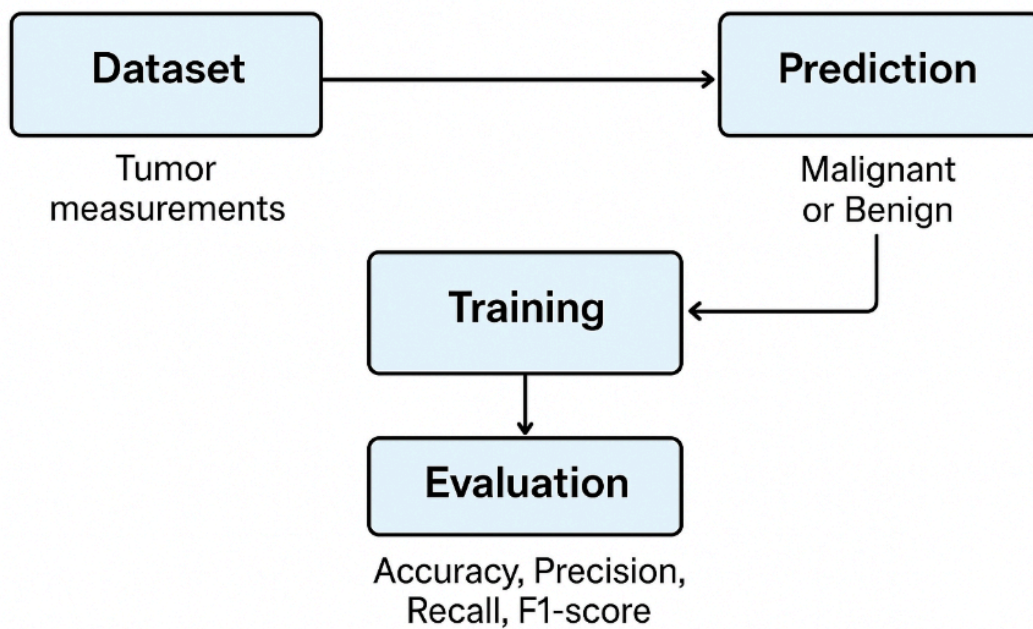
predict categories cat dog benign malignant 0, 1, 2

small number of possible outputs

Two or more inputs



Breast Cancer Classification



1. What is Supervised Learning?

- In **supervised learning**, you train an algorithm using a dataset that contains **inputs** (X) and **correct outputs** (Y)
 - The algorithm **learns the mapping**: $X \rightarrow Y$
 - The training data is **“labeled”** — meaning, for each example, you know the correct answer.
 - After training, the algorithm can predict Y for new, unseen X.
-

2. Two Main Types of Supervised Learning

A. Regression

- Predicts **continuous numeric values**.
 - Output can be *any number* from an **infinite range**.
 - Example: Predicting **house prices**, **temperature**, or **height**.
 - **Key point**: Outputs are real numbers like 15.3, 250000, 0.75, etc.
-

B. Classification

- Predicts **categories** (a limited, finite set of values).
 - Example in video: **Breast cancer detection**
 - Input: Patient's medical data (e.g., tumor size, age, cell shape)
 - Output:
 - **0** → **Benign** (not dangerous)
 - **1** → **Malignant** (dangerous)
 - **Key difference from regression**:
 - Regression can output **0.5**, **1.7**, and so on.
 - Classification outputs **are limited to predefined categories** (such as {0, 1, 2} or {cat, dog}).
-

3. Classification Details

- **Binary classification** → Two categories (e.g., benign/malignant).

- **Multi-class classification** → More than two categories (e.g., cancer type 0, 1, or 2).
 - Categories can be:
 - Numbers (0, 1, 2) — but they represent **labels**, not quantities.
 - Words (cat, dog, horse).
-

4. Example: Tumor Detection

- **One input feature:** Tumor size.
 - **Plotting data:**
 - 0 → benign tumors
 - 1 → malignant tumors
 - New patient's tumor size → Predict category.
 - **Two input features:** Tumor size + patient's age.
 - The algorithm finds a **boundary line** that separates benign from malignant cases.
 - If the new data point falls on one side of the line, → benign, other side → malignant.
 - In real-world applications, many additional features are utilized (such as cell thickness and shape).
-

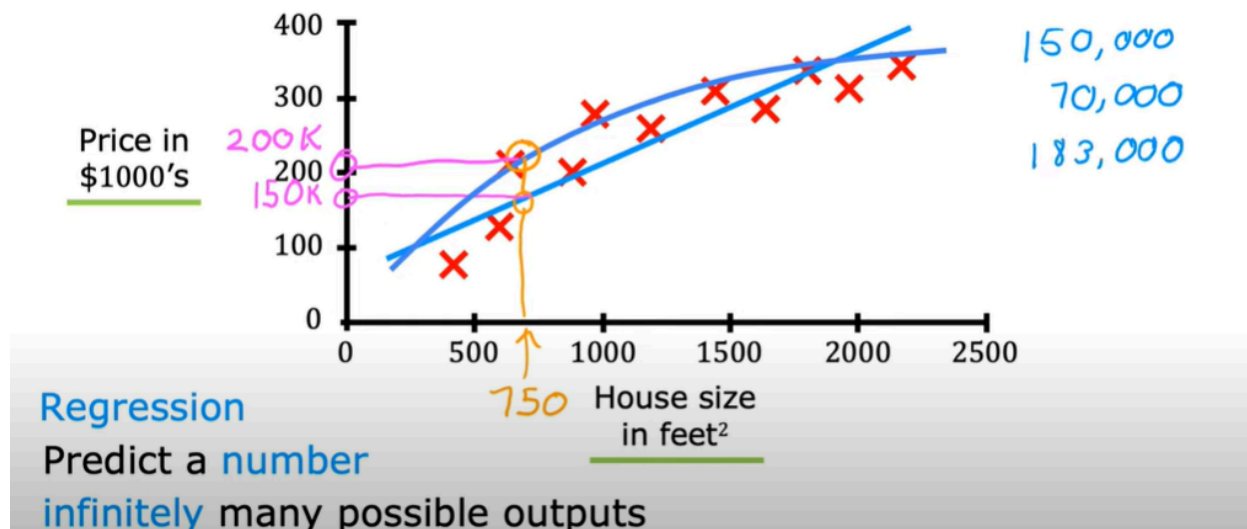
Regression Example:

Practical Example 2 (Regression)

Problem: Predicting car resale price

- **Input:** Car age, mileage, brand, fuel type
- **Label:** Price in dollars
- **Algorithm:** Linear Regression, Random Forest, etc.
- **Result:** Given details of a used car → Model predicts selling price.

Regression: Housing price prediction



Real-life examples:

1. What is Regression in Supervised Learning?

- In regression, the **output (label)** is **continuous** (numbers), not categories.
- The model learns the relationship between input variables (**features**) and a continuous target variable.
- Example: Predicting **house prices** based on square footage, number of rooms, and location.

2. Real-Life Examples

- Predict stock prices
- Estimate car fuel efficiency
- Forecast temperature
- Predict salary based on years of experience

3. Steps in Python

We'll use the **scikit-learn** library with a simple **Linear Regression** example.

Step 1 – Install libraries (if needed)

```
bash
```

```
pip install scikit-learn pandas matplotlib
```

Step 2 – Import libraries

```
python
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
```

Step 3 – Example dataset

We'll create a small dataset of **Years of Experience** vs **Salary**.

```
python 📄 Copy  
  
# Sample data  
data = {  
    'YearsExperience': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],  
    'Salary': [35000, 40000, 45000, 50000, 55000, 60000, 65000, 70000, 75000, 80000]  
}  
  
df = pd.DataFrame(data)  
print(df)
```

Step 4 – Split data

```
python 📄 Copy  
  
X = df[['YearsExperience']] # Feature  
y = df['Salary'] # Target  
  
# Split into 80% train, 20% test  
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Step 5 – Train the model

python

```
model = LinearRegression()  
model.fit(X_train, y_train)
```

Step 6 – Make predictions

python

```
y_pred = model.predict(X_test)  
  
print("Predictions:", y_pred)
```

Step 7 – Evaluate the model

python

```
mse = mean_squared_error(y_test, y_pred)  
r2 = r2_score(y_test, y_pred)  
  
print("Mean Squared Error:", mse)  
print("R2 Score:", r2)
```

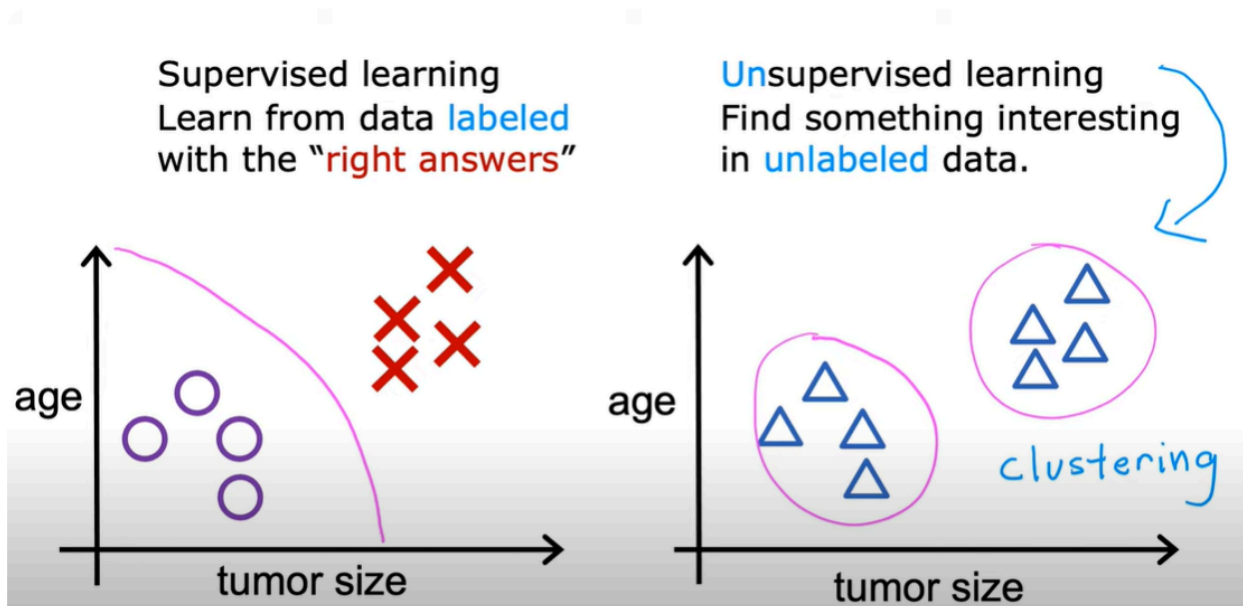


Step 8 – Visualize

python

```
plt.scatter(X, y, color='blue', label="Actual Data")
plt.plot(X, model.predict(X), color='red', label="Regression Line")
plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.legend()
plt.show()
```

What is Unsupervised Learning



Clustering: Google news

➤ Giant panda gives birth to rare twin cubs at Japan's oldest zoo

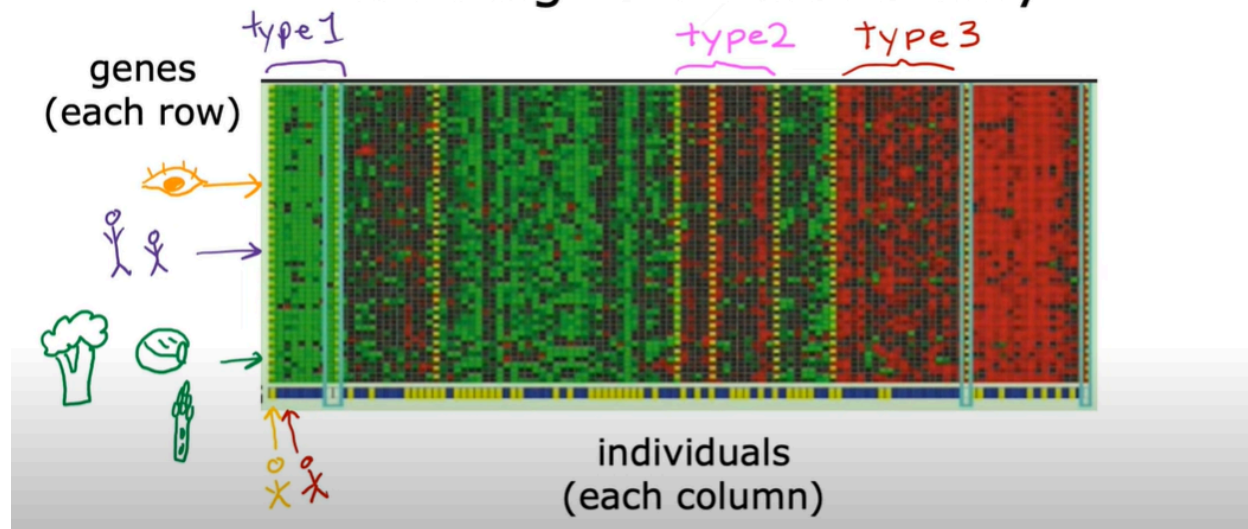
USA TODAY · 6 hours ago



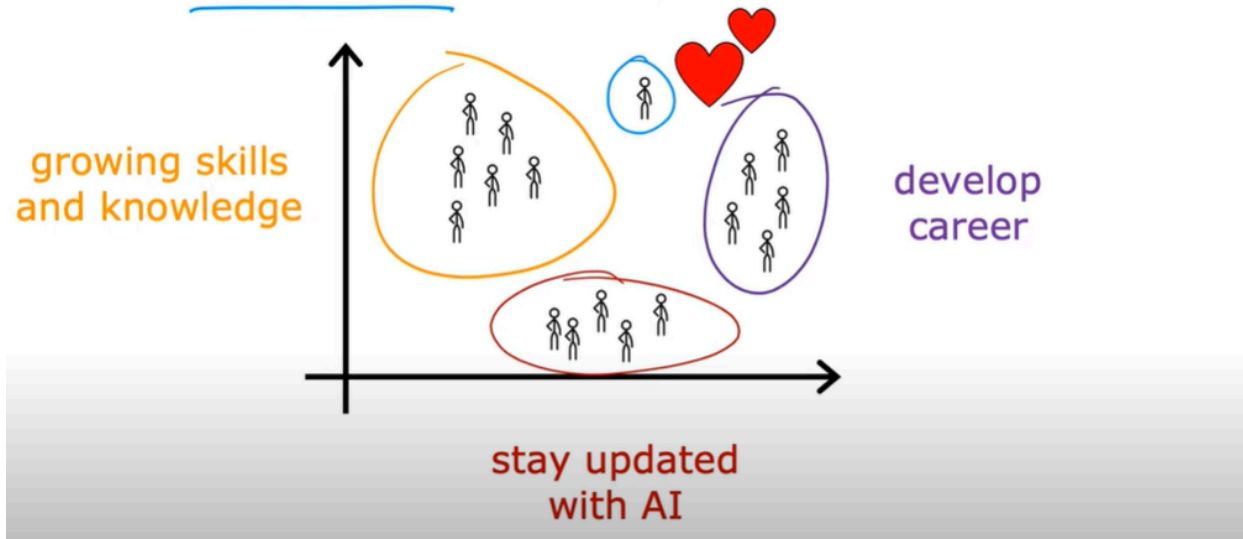
- Giant panda gives birth to twin cubs at Japan's oldest zoo
- Giant panda gives birth to twin cubs at Tokyo's Ueno Zoo
- A Joyful Surprise at Japan's Oldest Zoo: The Birth of Twin Pandas
- Twin Panda Cubs Born at Tokyo's Ueno Zoo

[View Full Coverage](#)

Clustering: DNA microarray



Clustering: Grouping customers



Key Points: Appendix

Machine Learning terminology

Definition

- **Arthur Samuel** defined machine learning as:
"The field of study that gives computers the ability to learn without being explicitly

programmed."

- Samuel developed a **checkers-playing program (ML Program)** in the 1950s.
 - He wasn't a great checkers player himself.
 - The program learned by playing **tens of thousands of games against itself**.
 - It analyzed board positions that led to **wins** and **losses**.
 - Over time, it became **better than Samuel himself**.
-

Core Idea

- The more **experience** a machine learning system gains (e.g., the more games it plays), the better it becomes.
 - Machine learning systems **improve performance by learning from data** instead of following fixed rules.
-

Two Main Types of Machine Learning

1. **Supervised Learning**
 - Most commonly used in real-world applications.
 - It will be the main focus in the early parts of the course.
 2. **Unsupervised Learning**
 - Introduced later, along with **recommender systems** and **reinforcement learning**.
-

Practical Application Focus

- Learning algorithms are like **tools**—important to know **how to use them effectively**, not just to have them.
- Many teams waste months on wrong approaches due to poor application of tools.
- This course emphasizes **best practices** for developing valuable ML systems.

1. What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence (AI) where computers learn from **data** to make predictions or decisions without being explicitly programmed for each task.

- Instead of giving the computer step-by-step instructions, we give it **examples**, and it **learns patterns** from them.

Example:

- If we want a computer to recognize cats in photos, we don't hardcode "cats have whiskers, tails, ears..." — instead, we show it thousands of labeled pictures ("cat" / "not cat") and let it learn patterns.
-

2. What is Supervised Learning?

Supervised Learning is the most common type of ML.

It works like a **teacher-student** model:

- The "teacher" gives the model examples with **both input (X)** and the **correct answer/output (Y)**.
- The model learns the relationship $X \rightarrow Y$ so it can predict Y **for new, unseen X**.

Key characteristics:

- **Labeled data** is required.
- The goal is to make predictions for new inputs.

Examples of Supervised Learning

Input (X)	Output (Y)	Application
Email text	Spam / Not Spam	Spam filter
Audio clip	Text transcript	Speech recognition
English sentence	Spanish sentence	Machine translation
Image of road	Steering direction	Self-driving cars
Product photo	Defective / Not defective	Quality inspection
House size	Price in \$	Real estate price prediction

3. How it Works (Step-by-Step)

1. **Collect labeled data** (input-output pairs).
Example: House size $\rightarrow$ House price.
2. **Train the model** by showing these pairs repeatedly.
3. **The model learns patterns** between X and Y.
4. **Test the model** on new, unseen inputs.

5. Use it in real-world predictions.
-

4. Types of Supervised Learning

1. **Regression** → Predicting a continuous number.
Example: House price prediction.
 - Input: 750 sq. ft.
 - Output: \$200,000 (could be any number).
2. **Classification** → Predicting a category or class.
Example: Email spam detection.
 - Input: Email text.
 - Output: "Spam" or "Not Spam" (fixed set of labels).

5. Practical Example – Predicting House Prices

Let's say:

- X = Size of house (sq. ft.)
- Y = Price of house

Data:

Size (sq. ft.)	Price (\$)
500	120,000
1000	200,000
1500	270,000

Steps:

1. Plot these points on a graph.
2. Fit a model (straight line or curve).
3. For a **new house** of 750 sq. ft., the model predicts a price.

Why supervised?

Because we gave the algorithm both **inputs (size)** and **correct answers (price)** for training.

✅ **Summary:**

- **Machine Learning** = Teaching computers from data.
- **Supervised Learning** = Teaching computers with labeled examples (input-output pairs).
- **Regression** predicts numbers; **Classification** predicts categories.

What does that mean? Fit a model (straight line or curve)?

1. Imagine your data

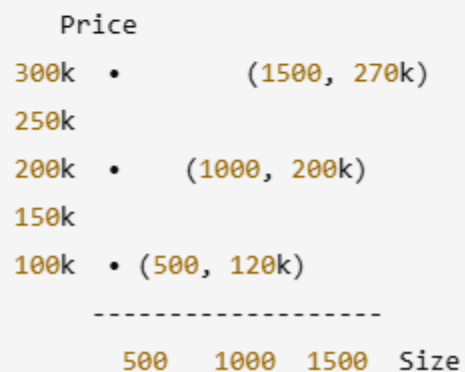
Example:

House size (X) vs. Price (Y)

Size (sq. ft.)	Price (\$)
500	120,000
1000	200,000
1500	270,000

If you plot these points on a graph, they might look like this:

SCSS



2. "Fitting a model" means:

We draw a **line** or **curve** through these points that:

- **Best matches** the trend of the data
 - Minimizes the prediction error for all points
-

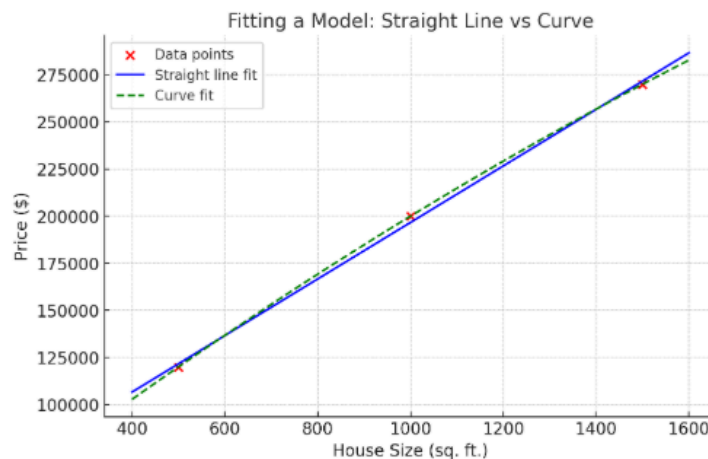
3. Why a straight line or curve?

- If the relationship between X and Y is roughly constant, we use a **straight line** (Linear Regression).
 - Formula: $\text{Price} = m \times \text{Size} + b$
Example: $\text{Price} = 150 \times \text{Size} + 50,000$
 - If the relationship is more complex (e.g., speeds up or slows down), we might use a **curve** (Polynomial Regression, Exponential, etc.).
 - Example: Car speed vs. fuel consumption might need a U-shaped curve.
-

4. Practical analogy

Think of it like this:

- You have scattered dots on paper (your data).
- You take a pen and try to draw one smooth line that goes as close as possible to **all** the dots.
- That line is your **model**.
- Once you have it, you can use it to predict new points.



"Fit a model (straight line or curve)" means finding a mathematical function that best represents the relationship between your input data (e.g., house size) and your output data (e.g., price).

- **Straight line fit** → assumes the relationship is linear (price changes at a constant rate with size).
- **Curve fit** → assumes the relationship is non-linear (price changes at varying rates with size).

In machine learning, this "fitting" process is how the algorithm learns from the data to make predictions. `</>`

Supervised learning

Learns from being given "right answers"

Regression

Predict a **number**

infinitely many possible outputs

Classification

predict **categories**

small number of possible outputs

Role of Model in Machine Learning

In machine learning, the **main job of a model** is to **learn patterns from data** so it can make predictions or decisions on new, unseen data.

Concept

- The **model** is essentially the mathematical or algorithmic representation of the problem.
 - During **training**, it looks for patterns in the input features and their relationship to the output labels.
 - Once trained, it can **generalize** those patterns to make predictions on new data.
-

Example Roles of a Model

1. **Classification** – Assigns an input to one of several categories.
 - Example: A breast cancer classifier that predicts whether a tumor is *benign* or *malignant*.
 - Model types: Logistic Regression, Random Forest, Support Vector Machine.
2. **Regression** – Predicts a continuous numerical value.
 - Example: Predicting house prices based on size, location, and features.
 - Model types: Linear Regression, Gradient Boosting Regressor.
3. **Clustering** – Groups similar data points together without predefined labels.
 - Example: Customer segmentation for targeted marketing.
 - Model types: K-Means, DBSCAN.
4. **Recommendation** – Suggests items to users based on preferences and behavior.
 - Example: Netflix suggesting movies based on your viewing history.
 - Model types: Collaborative Filtering, Matrix Factorization.

✅ In short:

A machine learning model's job is **pattern learning + prediction**.

Think of it like a student:

- **Training phase** → the student studies examples and answers.
 - **Prediction phase** → the student answers new questions using what they've learned.
-

Unsupervised Learning Overview

- The second most widely used form of ML is after supervised learning.
- **No output labels (y)** are provided; the algorithm must find **patterns** or **structure** in the data by itself.

- Goal: discover interesting **groupings**, **patterns**, or **relationships** in the data.
 - Called "**unsupervised**" because the algorithm is not guided toward a predefined "right" answer.
-

Example 1 – Clustering in Tumor Data

- Data: patient age & tumor size, but **no labels** (benign/malignant).
 - An algorithm may group data into **clusters** based on similarity.
 - **Clustering** = placing **unlabeled data** into groups.
-

Applications of Clustering

Google News

- Group hundreds of thousands of daily articles into related story clusters.
- Finds common **keywords** (e.g., "panda," "twins," "zoo") automatically.
- No human explicitly tells it which words to look for; the algorithm adapts daily to new topics.

DNA Microarray Data

- Each column = one person's DNA expression data.
- Each row = one gene (e.g., eye color, height, vegetable preference).
- Colors show gene activity levels.
- Clustering groups individuals into **genetic types** (Type 1, Type 2, Type 3).
- No prior information about the **number/types** of groups is given to the algorithm.

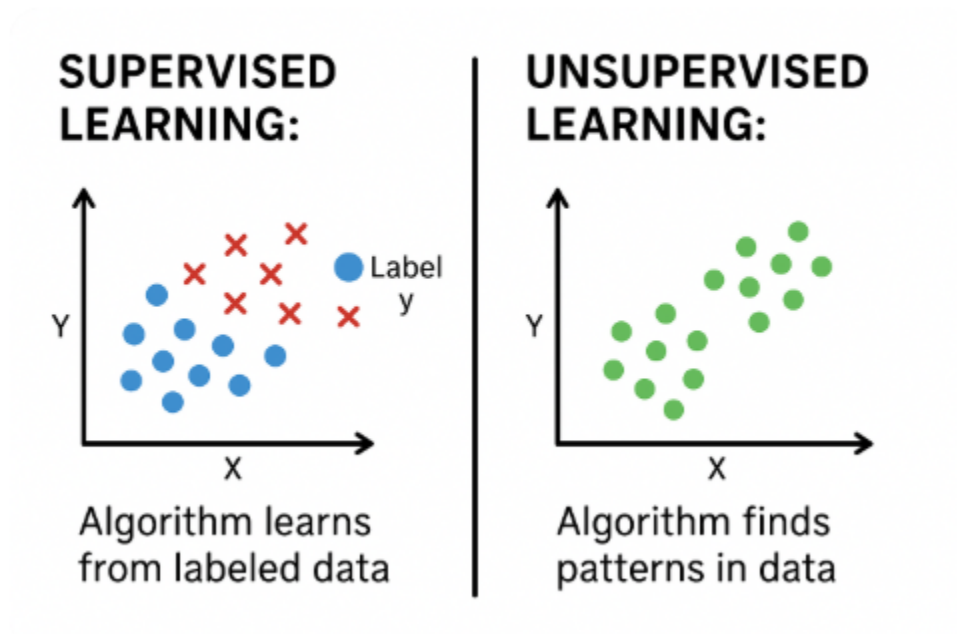
Customer Segmentation

- Companies use clustering to group customers into **market segments**.
 - Example: DeepLearning.AI community clustering identified groups with different motivations:
 1. Skill growth & knowledge seekers.
 2. Career advancement seekers.
 3. AI impacts observers in their field.
 - Helps tailor services and communication for each group.
-

Key Takeaways

- **Clustering** is = most common unsupervised learning method.
- Works without labels, automatically finding groups in the data.
- Used in:

- News article grouping
- Genetic data analysis
- Market/customer segmentation
- Other types of unsupervised learning also exist beyond clustering.

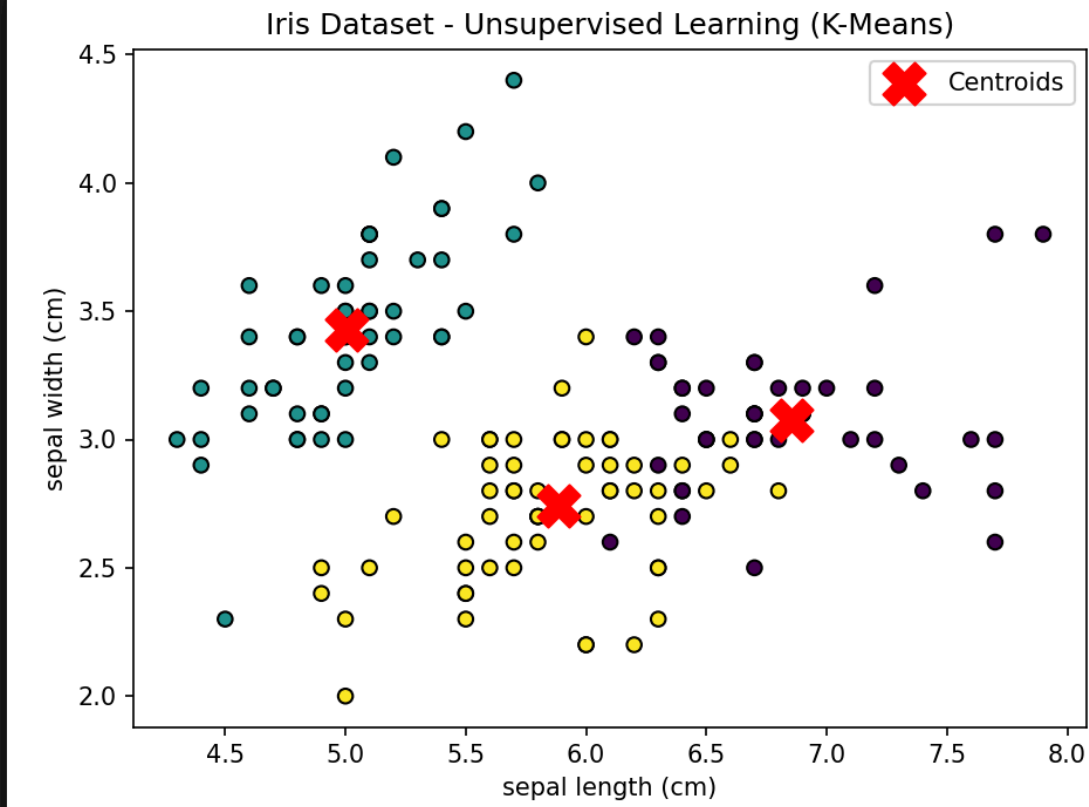


Iris Dataset - unsupervised Learning (K-means)

critical but still good practice.

Cluster labels are not stable. The same dataset can be grouped differently depending on the initial centroid values.

Quality
of
the
clusters



• Source: Iris dataset (150 flower samples).

1. Dataset Context

- **Source:** Iris dataset (150 flower samples).
 - **Features used in plot:**
 - **X-axis:** Sepal length (cm)
 - **Y-axis:** Sepal width (cm)
 - Even though each flower also has petal measurements, we only plot these two dimensions so it's visible in 2D.
-

2. Data Points (Colored Circles)

- **Each dot** = one flower from the dataset.
 - **Color:** Assigned by **K-Means cluster ID** (0, 1, 2).
 - Yellow, teal, and purple here represent the three clusters.
 - These colors are **not species labels** — K-Means picks cluster numbers arbitrarily.
 - **Black edges** on circles make points easier to see against each other.
-

3. Clusters & Separation

- K-Means grouped flowers into 3 clusters based on sepal length and width similarity.
- In reality, the Iris dataset has 3 species:
 - *Iris setosa*
 - *Iris versicolor*
 - *Iris virginica*
- Because this is **unsupervised**, the algorithm doesn't know their names — it just groups them by distance in feature space.

4. Red “X” Markers (Centroids)

- **Big red X's** = cluster centers (mean position of points in each group).
 - Each centroid's coordinates are the **average sepal length and sepal width** for flowers in that cluster.
 - K-Means iteratively moves these centroids until they stabilize — meaning minimal distance from each point to its nearest centroid.
-

5. Observations

- The **left/top cluster** (teal) is fairly separated from others — likely corresponds to *Iris setosa*, which is distinctly smaller sepal length and larger sepal width.
 - The **bottom middle cluster** (yellow) and **right cluster** (purple) are closer and overlap more — these likely represent *versicolor* and *virginica*, which are harder to separate using just sepal measurements.
 - If we used **all 4 features** (including petal length/width), the clusters would separate more cleanly.
-

6. Limitations

- This plot only shows 2 features, so some separation is lost.
- K-Means assumes clusters are spherical and equally sized — not always the case in real-world data.
- Cluster IDs (0, 1, 2) don't correspond to actual species IDs without matching afterward.